

Probabilistic Graphical Models

CSE 5830

Representation

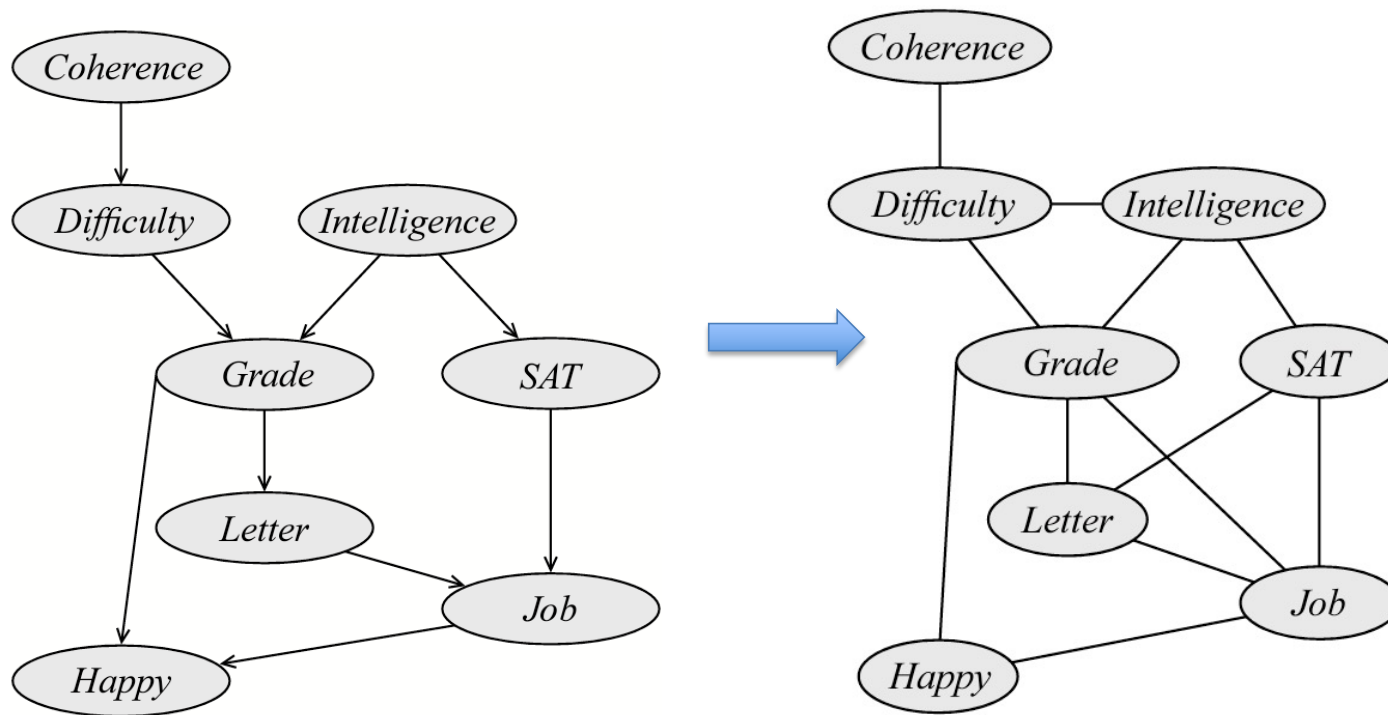
Lecture 7

**Markov Network,
Templates**

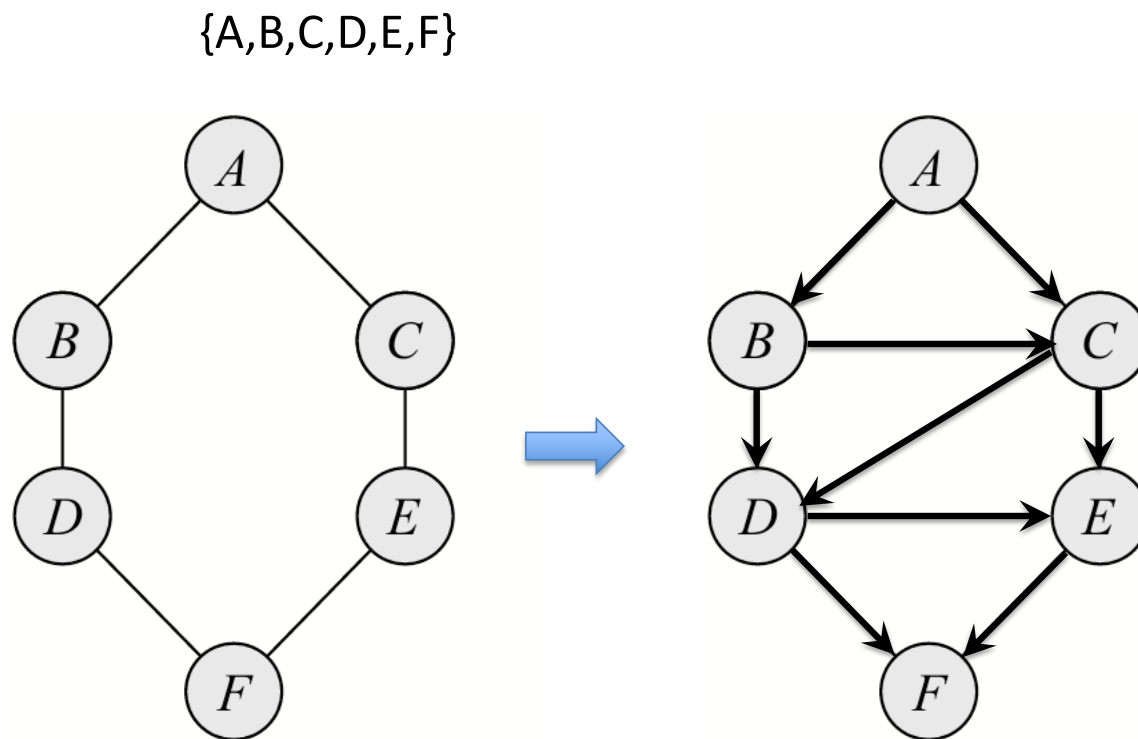
Goals for Today

- Markov network (undirected graph)
 - Transferring between BNs and MNs
- Templates
 - Temporal Models
 - 2-time-slice Bayesian Network
 - Dynamic Bayesian networks
 - Hidden Markov Model
 - Plate models

Example of BN $\rightarrow$ MN (Review)

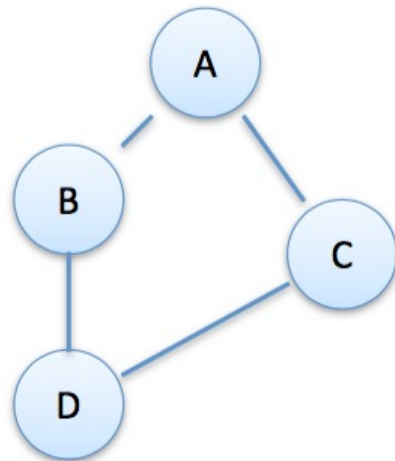


Example of $MN \rightarrow BN$ (Review)

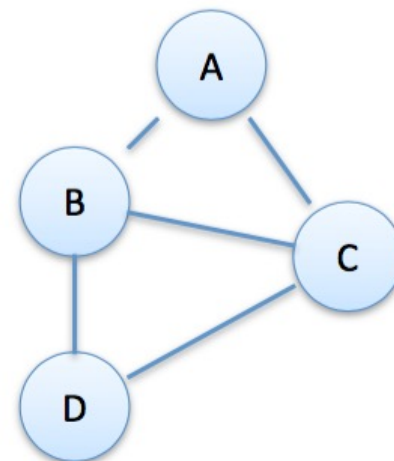


Chordal Graphs

- Graph whose minimal cycles are not longer than 3.



Not Chordal Graph



Chordal Graph

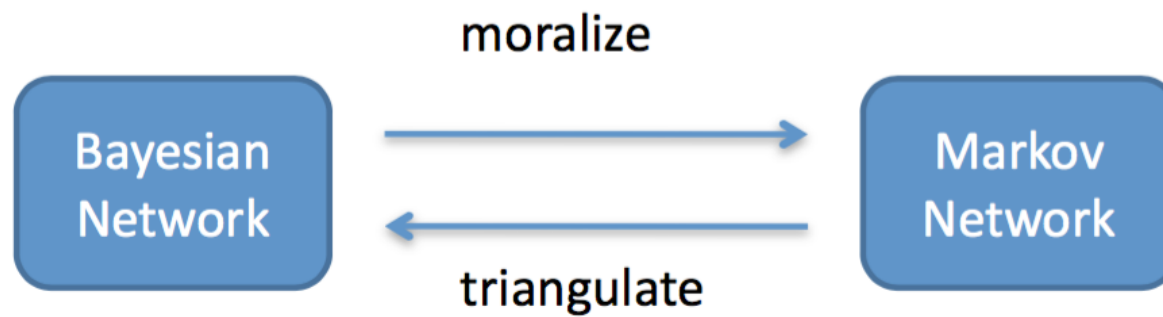
Chordal graph for $MN \rightarrow BN$

- The Bayesian network we create cannot have any immoralities!
- If G is a minimal I-map Bayesian network for Markov network H , then G has no immoralities.
 - It is **chordal**,
since any loop of length ≥ 4 in a Bayesian network graph must have immoralities.

Markov Network to Bayesian Network

- Conversion from MN to BN requires triangulation
 - May lose some independence information.
 - May involve a lot of additional edges.
 - Different orderings of chain rule may yield different numbers of additional edges.

Transferring between BNs and MNs

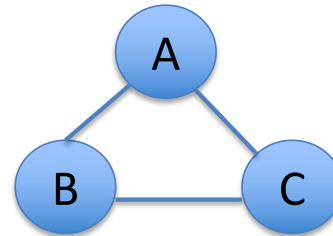


Perfect transformation between BN and MN

- Bayesian network/Markov network conversion can lead to addition of edges and loss of independence information
- Is there a subset of distributions that can be captured *perfectly* in both models?
 - Yes! Undirected chordal graphs.

Factor Graph

- Graphical models
 - Bayesian Network
 - Markov Network
 - Factor Graph



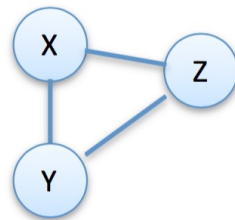
- Factor graph is for representation of factorization of a distribution
 - Alternative representation for parameterization of a Markov Network
 - typically equivalent to independence statements amongst (sets of) variables in the distribution.

Representing factorization in Graphs

- Belief Network, Bayesian Network
 - Each factor is a conditional distribution
 - Generative models, AI, statistics
 - Corresponds to a DAG
- Markov Network, Undirected graphs
 - Each factor corresponds to a potential (non negative function)
 - Related to the strength of relationship between variables but not directly related to dependence
 - Useful for collective phenomena such as image processing
- Factor Graph
 - A barebones representation of the factorization of a distribution
 - Often used for efficient computation and deriving message passing algorithms.¹²

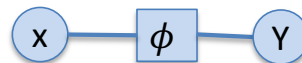
Factor graph

- Makes the structure of the Gibbs distribution explicit

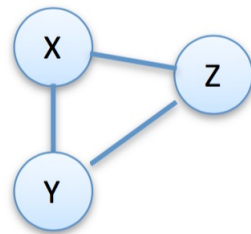


Which Gibbs dis.?

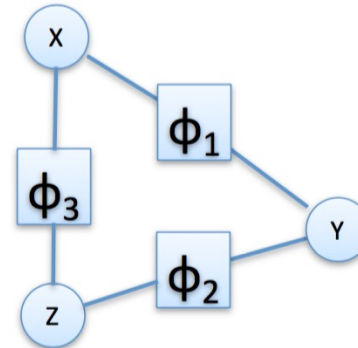
- Contains two types of node
 - Random variables
 - Factors over the variables
- Contains edges between variable nodes and factor nodes.
- Each factor node V_ϕ is associated with one factor ϕ , which its scope is neighboring variables of V_ϕ



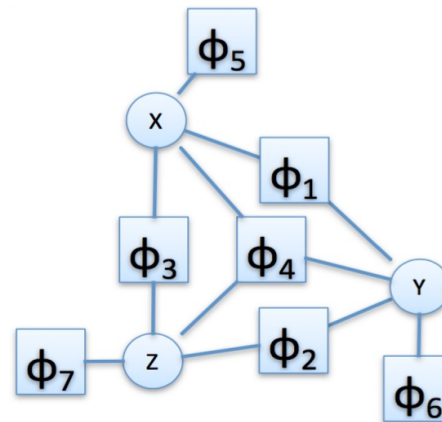
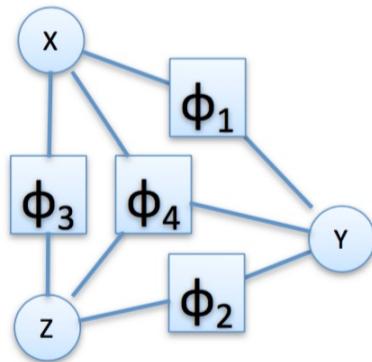
Factor graph



pairwise Markov network



all cliques



Template based representation

Template models

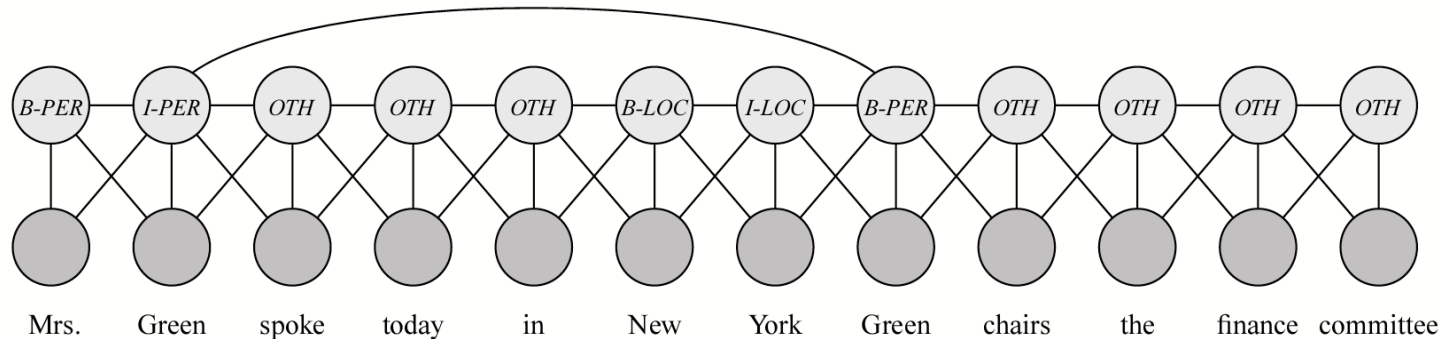
- A PGM specifies a joint distribution mostly over a fixed set of random variables
- The distribution is used in a variety of different situations
- Different situations share the same general structure
- How can we compactly represent the distribution over all situations?

Templates

Model representation

- Lots of commonalities
 - Random variable shared structures and/or dependency models
- Build a single compact model that provides a template for an entire class of distributions → templates
- Reuse of parameters and structure

Example: NLP sequence model

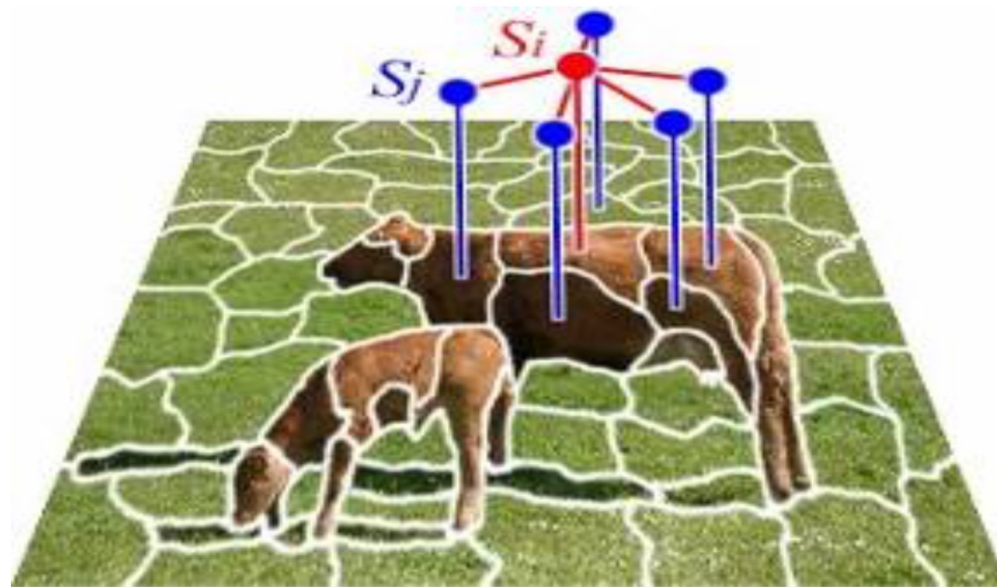


KEY

<i>B-PER</i>	Begin person name	<i>I-LOC</i>	Within location name
<i>I-PER</i>	Within person name	<i>OTH</i>	Not an entity
<i>B-LOC</i>	Begin location name		

- Same model, reuse of parameters
- Model is independent from the place in the sequence

Example: Image segmentation

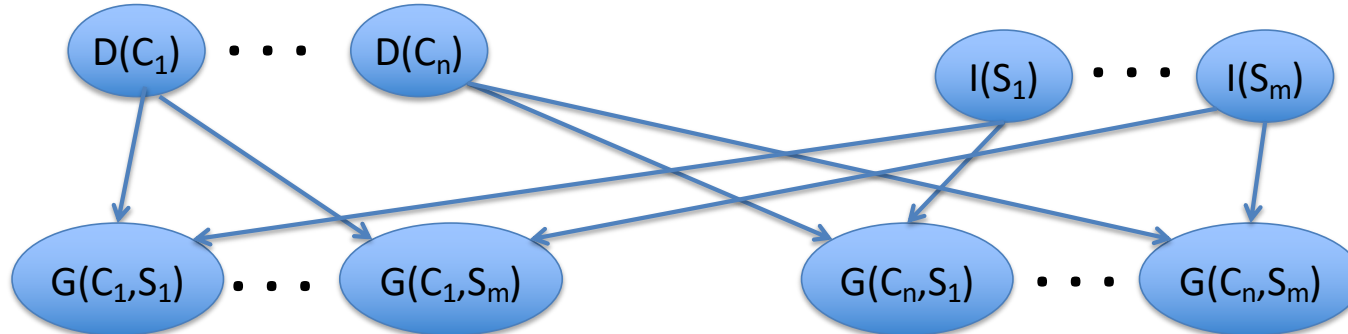
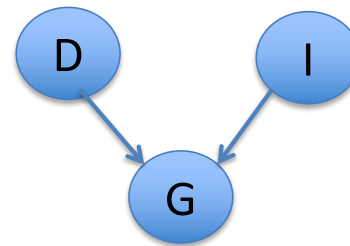


Daphne Koller

- Same model for all super pixels
- Relation between class labels and pixel features is the same
- The same model across images

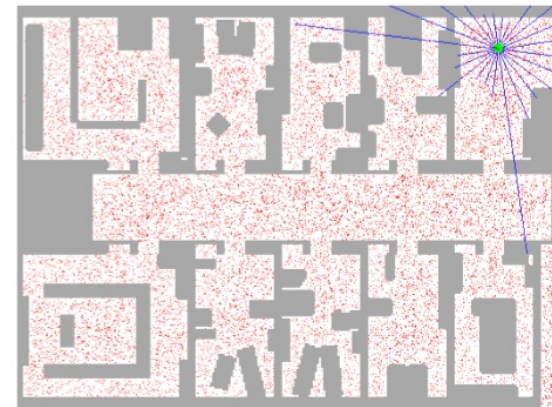
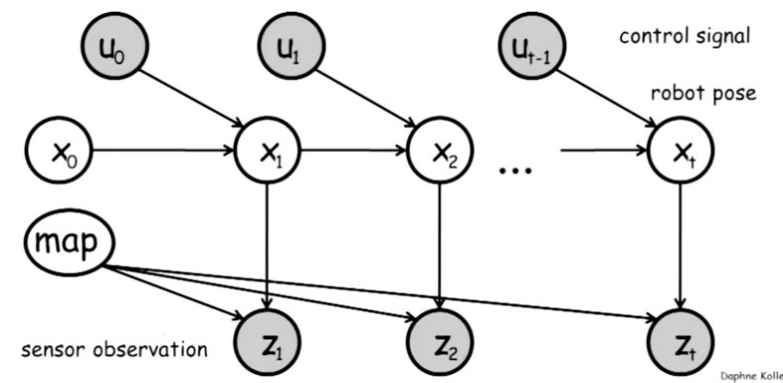
Example: Students' grades

What about n courses
and m students?



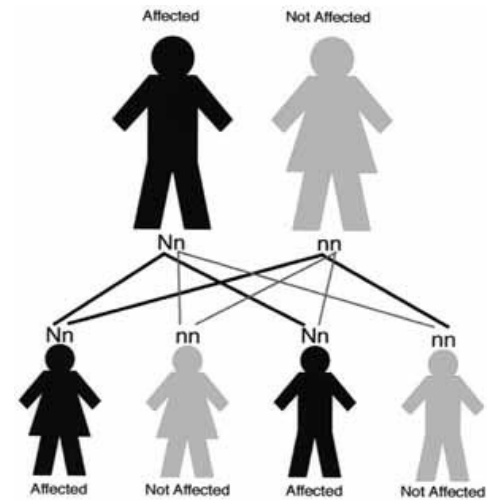
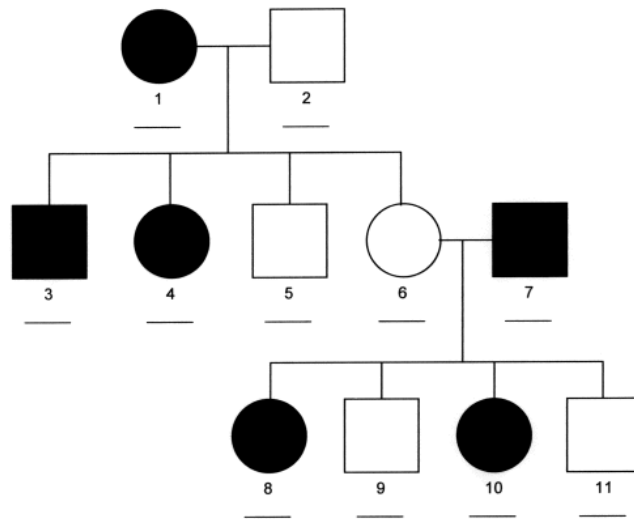
Sharing structures and parameters

Example: Robotic localization



Same model for moving in time 1 and time 2

Example: Genetic inheritance



Same structure for all people

Template Variables

- Template variable $X(U_1, \dots, U_k)$ is instantiated (duplicated) multiple times
 - X represents a family of random variables
 - $U_1, \dots, U_k$ are entities over which the template is instantiated
- Examples:
 - Location(t)
 - Genotype(person), Phenotype(person)
 - Label(pixel)
 - Difficulty(course), Intelligence(student), Grade(course, student)

Template Models

- Temporal models
 - Capturing properties of the system dynamic
 - Robot location
 - Dynamic Bayesian networks
- Object-relational models
 - no temporal
 - Students grades, genetic inheritance
 - Plate models

Temporal Models

Modeling dynamic setting

– Distribution over systems $\rightarrow$ state changes over time

- $X_i^{(t)}$: instantiation of the X_i at time t
 - Template variable
- For a set of variable $\mathbf{X}$,

$$\{\mathbf{X}^{(t)} : t \in [t_1, t_2]\} \quad \mathbf{X}^{(t_1:t_2)} \quad (t_1 < t_2)$$

$\mathbf{X}^{(t_1:t_2)}$: an assignment

- Trajectory: an assignment to $X_i^{(t)}$ for each time t

Temporal models represent joint distribution over a trajectory

Distributions over trajectories

- Δ – time granularity
- $X^{(t)}$ – variable X at time $t \times \Delta$
- $X^{(0:t)} = \{X^{(0)} \dots, X^{(t)}\}, \quad (0 \leq t)$

We want to represent $P(X^{(0:t)})$ for any t

- In general probability over possible trajectory is very complex
- Make a series of simplifying assumptions
 - Markov assumption
 - Time invariance

Markov assumption

- Chain rule of probabilities

$$P(\mathcal{X}^{(0:T)}) = P(\mathcal{X}^{(0)}) \prod_{t=0}^{T-1} P(\mathcal{X}^{(t+1)} \mid \mathcal{X}^{(0:t)})$$

Assuming:

$$(\mathcal{X}^{(t+1)} \perp \mathcal{X}^{(0:(t-1))} \mid \mathcal{X}^{(t)})$$

Future is independent of past given present

 $P(\mathcal{X}^{(0)}, \mathcal{X}^{(1)}, \dots, \mathcal{X}^{(T)}) = P(\mathcal{X}^{(0)}) \prod_{t=0}^{T-1} P(\mathcal{X}^{(t+1)} \mid \mathcal{X}^{(t)}).$

Time Invariance

- Stationary Markovian dynamic systems
- Transition model for any $t \geq 0$

$$P(\mathcal{X}' | \mathcal{X})$$

$$P(\mathcal{X}^{(t+1)} = \xi' | \mathcal{X}^{(t)} = \xi) = P(\mathcal{X}' = \xi' | \mathcal{X} = \xi)$$

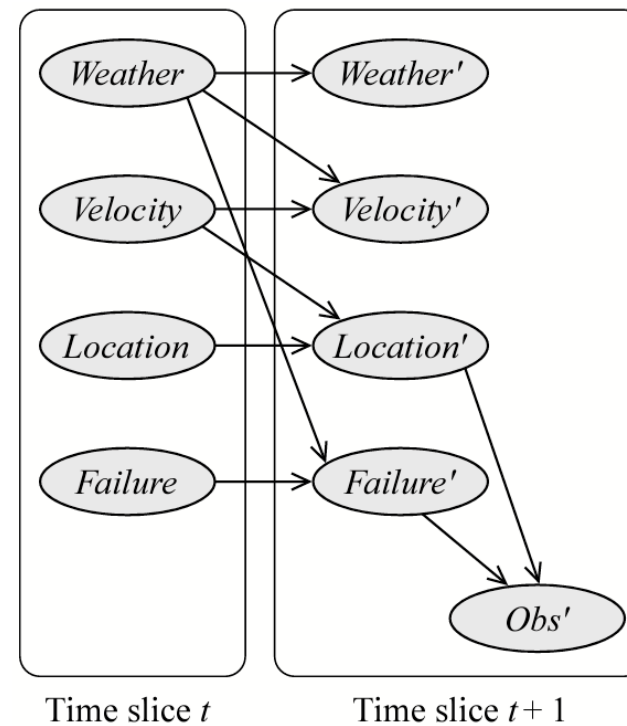
- The sequence of the event is important not the time of occurring
 - We only need two time instances
 - Conditional Bayesian network

Example: Template transition model

Example: modeling traffic

- We only need two time instances
- Conditional Bayesian network

$$P(W', V', L', F', O' | W, V, L, F)$$



(a) $\mathcal{B}_{\rightarrow}$

2-time-slice Bayesian Network

- A transition model (2TBN) over $X_1, \dots, X_n$ is specified as a BN fragment such that:
 - The nodes include $X_1', \dots, X_n'$ and a subset of $X_1, \dots, X_n$
 - Only the nodes $X_1', \dots, X_n'$ have parents and a CPD
- The 2TBN defines a conditional distribution

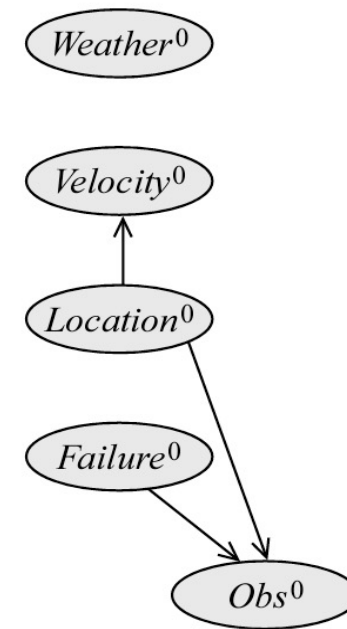
$$P(\mathcal{X}' \mid \mathcal{X}) = \prod_{i=1}^n P(X'_i \mid \text{Pa}_{X'_i})$$

Initial State Distribution

$$P(\mathcal{X}^{(0)}, \mathcal{X}^{(1)}, \dots, \mathcal{X}^{(T)}) = P(\mathcal{X}^{(0)}) \prod_{t=0}^{T-1} P(\mathcal{X}^{(t+1)} \mid \mathcal{X}^{(t)}).$$

We also need to specify the beginning states

$$P(W^{(0)}, V^{(0)}, L^{(0)}, F^{(0)}, O^{(0)})$$



Time slice 0

(b) $\mathcal{B}_0$

Dynamic Bayesian Network

- A dynamic Bayesian network (DBN) over $X_1, \dots, X_n$ is defined by a
 - 2TBN $\text{BN}_{\rightarrow}$ over $X_1, \dots, X_n$ (transition model)
 - a Bayesian network $\text{BN}^{(0)}$ over $X_1^{(0)}, \dots, X_n^{(0)}$
- Modeling using a pair of $(\text{BN}^{(0)}, \text{BN}_{\rightarrow})$
- Assumption
 - Markov independencies
 - Time invariance

Unrolled DBM, Ground Network

- For a trajectory over $0, \dots, T$ we define a ground (unrolled) network such that
 - The dependency model for $X_1^{(0)}, \dots, X_n^{(0)}$ is copied from $\text{BN}^{(0)}$
 - The dependency model for $X_1^{(t)}, \dots, X_n^{(t)}$ for all $t > 0$ is copied from $\text{BN}_{\rightarrow}$

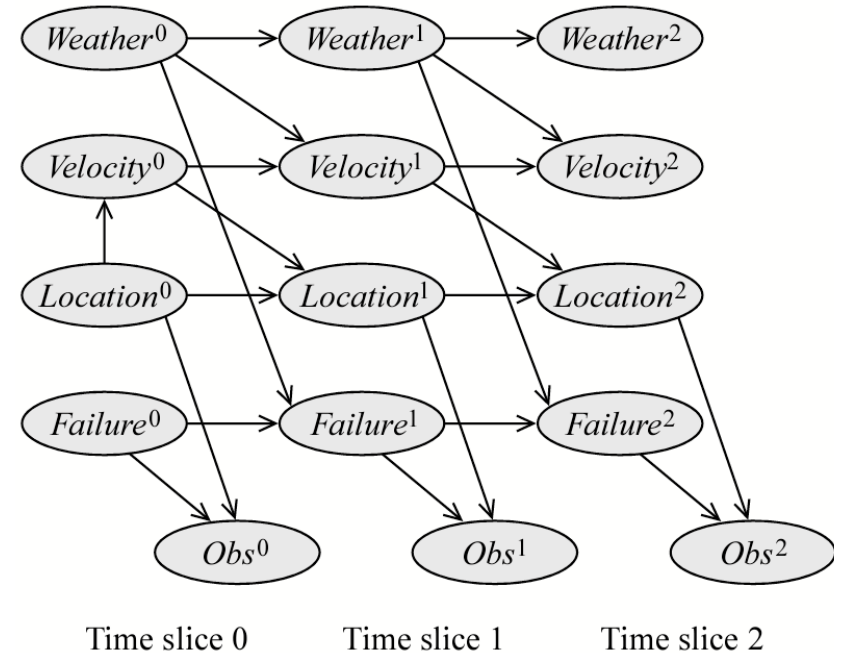
Example: Ground Bayesian networks

Unrolled DBN

- $Obs^{(i)}$, are all leaves
 - Observation variables
- Parents only in X'
 - State variables

We can see temporal systems as a state-observation model

- Separate states from observation, independency, for more efficient computation
- $P(X' | X)$ transition model
- $P(O | X)$ observation model



State-observation models

State model: Alternative way of representing temporal processes

- $\mathcal{S}^{(i)}$: a set of state variables
- Two independencies assumptions
 - Markovian independencies for state variables

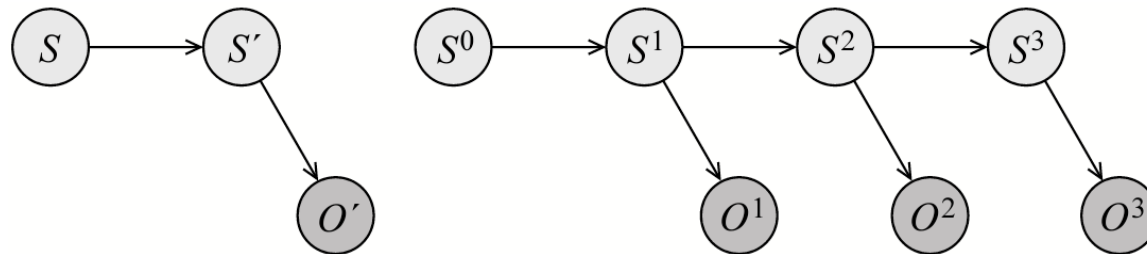
$$(\mathcal{S}^{(i+1)} \perp \mathcal{S}^{(0:(i-1))} | \mathcal{S}^i)$$

- Observation variables, $\mathcal{O}^{(i)}$, are conditionally independent of the entire state variables given the current state

$$(\mathcal{O}^{(i)} \perp \mathcal{S}^{(0:(i-1))}, \mathcal{S}^{(i+1:\infty)} | \mathcal{S}^i)$$

State-observation models

- Nodes are states, Leaves are Obs
- Weight of the edges represent transition probabilities $P(S' | S)$



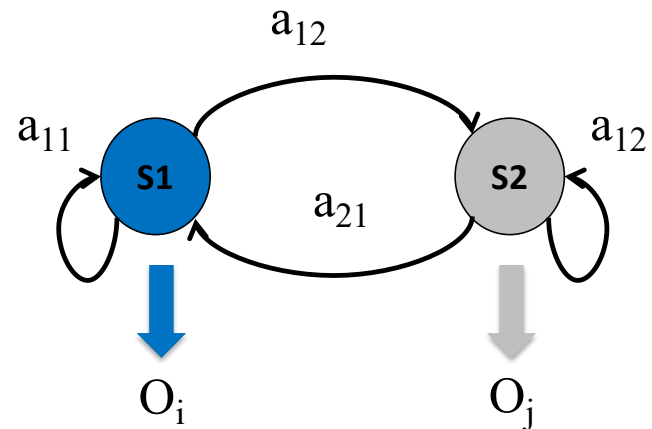
- Observations can occur in a separate process
 - Noisy sensors
- Separating system dynamic from the observation model

Hidden Markov Model (HMM)

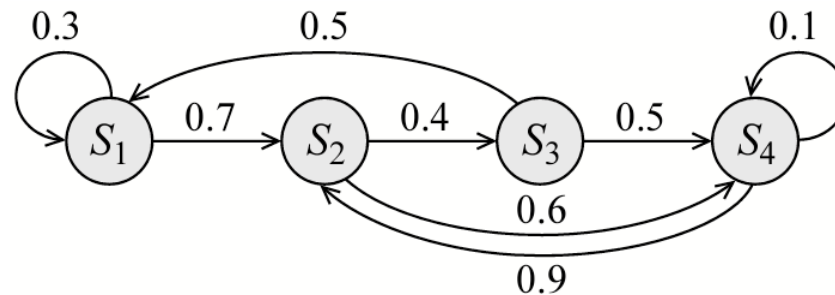
- Markov process with unobserved (hidden) states
- A state is not directly visible, but output, dependent on the state, is visible.
- Transition model

$$P(S' | S)$$

- Nodes are states
- Leaves are Obs
- Weight of the edges represent transition probabilities $P(S'|S)$



Hidden Markov Model (HMM)



Nodes are states

Weight of the edges are $P(S'|S)$

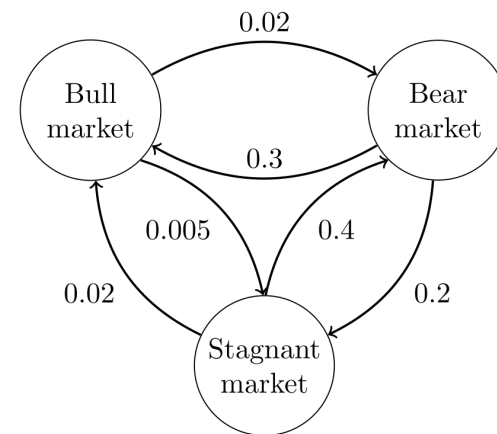
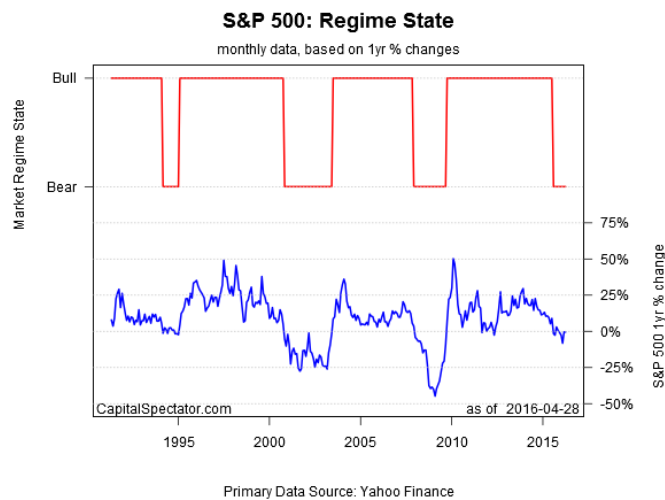
	s_1	s_2	s_3	s_4
s_1	0.3	0.7	0	0
s_2	0	0	0.4	0.6
s_3	0.5	0	0	0.5
s_4	0	0.9	0	0.1

HMM Applications

HMM is commonly used

- examples
 - Stock market
 - Robot localization
 - Speech recognition
 - Digital signal processing
 - Bioinformatics

Stock Market



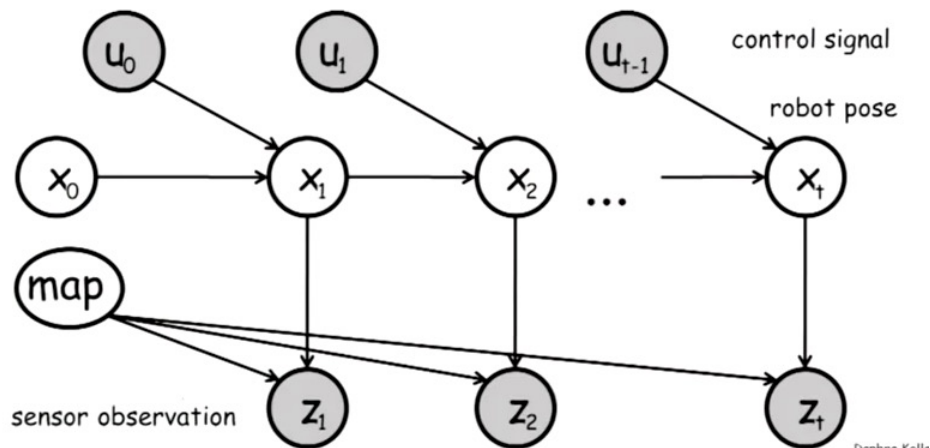
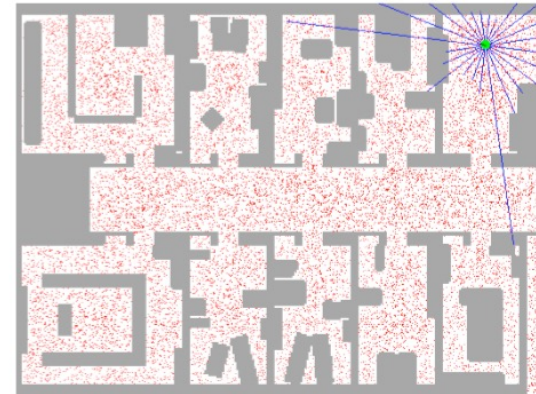
bull/bear market hidden Markov chain

- stock daily up/down observed
- depends on big market trend

Robotic localization

Robot pose Markov chain

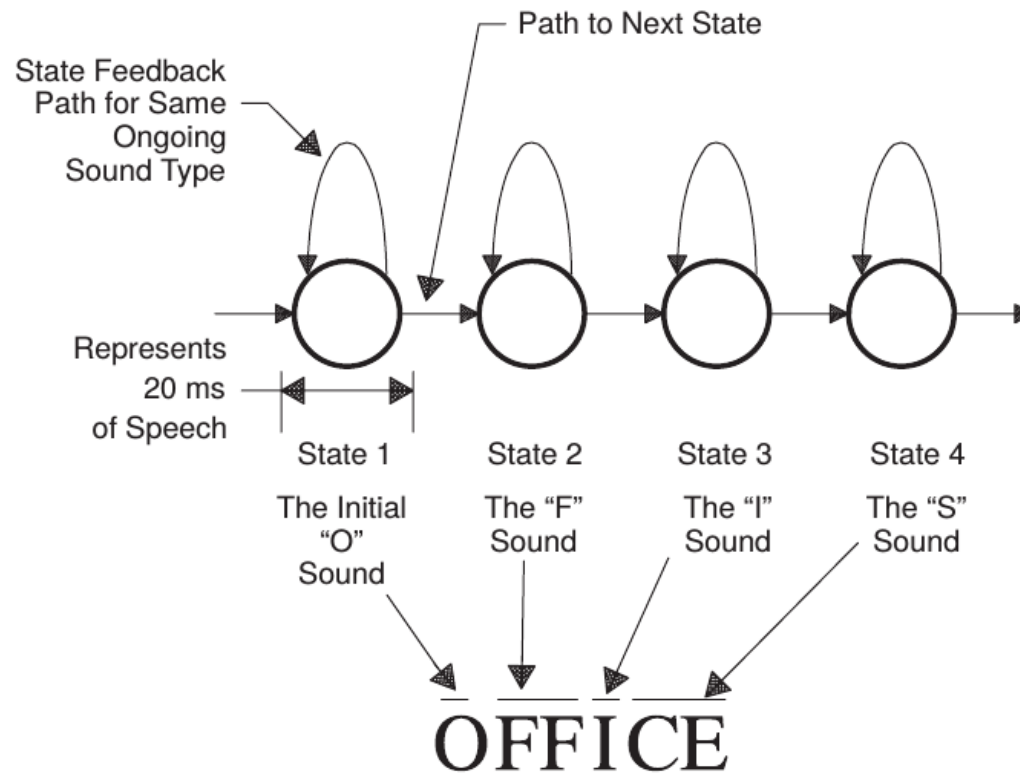
- sensors output observed
- depends on the robot pose



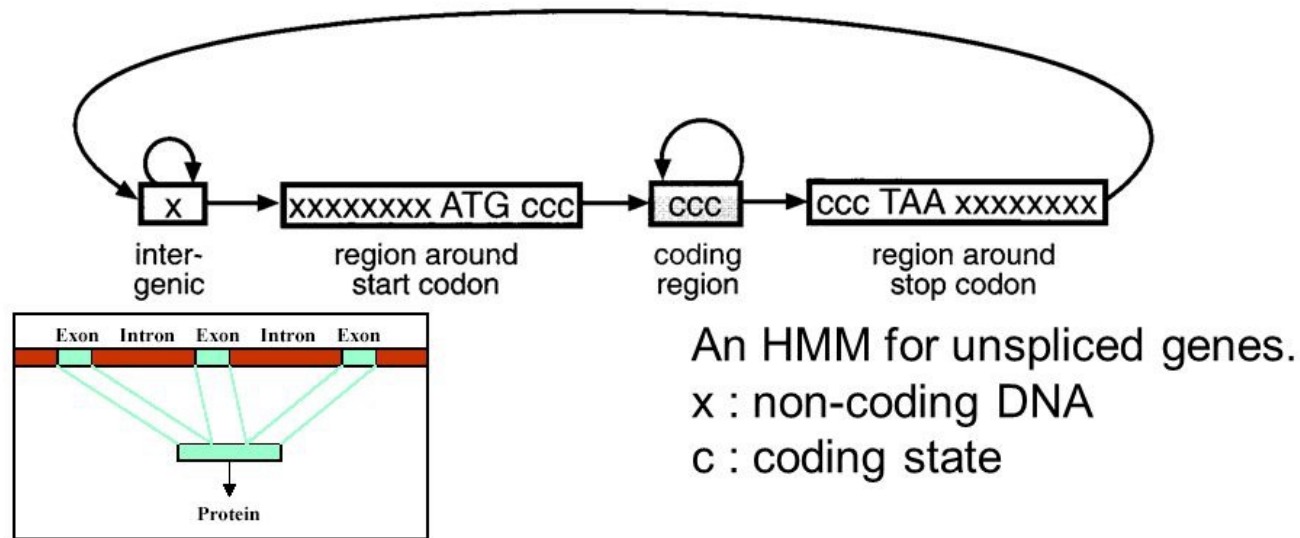
Speech recognition

Sentences & words
hidden Markov chain

- spoken sound
observed (heard)
- depends on the
words



Gene finding



Unspliced genes Markov chain

- sequence of nucleotide are observed
- genes are hidden
- depends on the sequence

Plate models

- Object relational frameworks (no temporal)
 - Template-based
- Object types called plates
 - Multiple objects share the same probabilistic model $\rightarrow$ plates
- Modeling repetition
 - Example:
Coin, $P(\text{head} = T)$

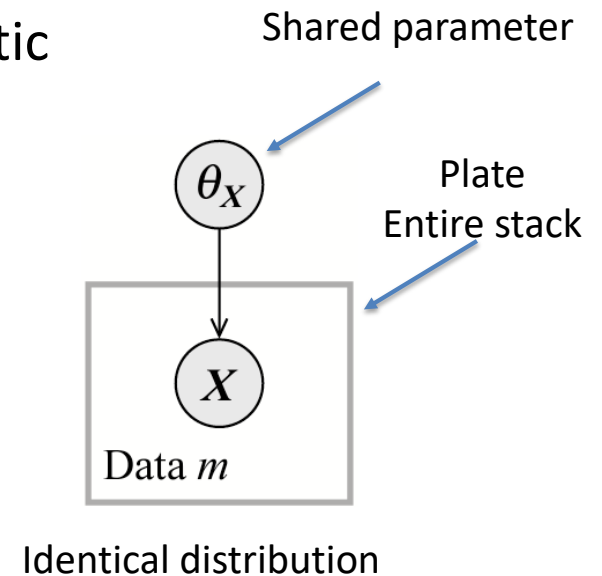


Plate models, example



We do not show common parameters

CPD for $P(I)$ and $P(G | I)$ is the same for all students

What about more complex system with multiple random variables that are connected?

→ Adding more plates

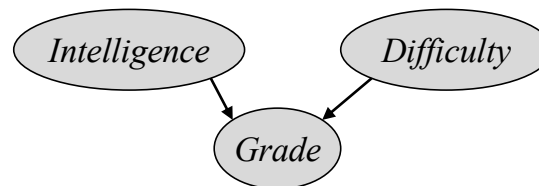
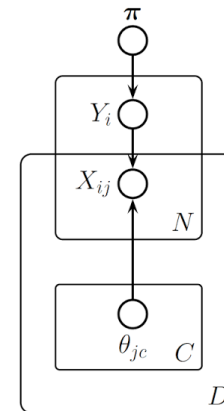
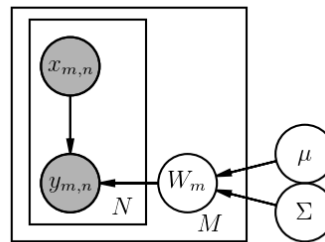


Plate models

Models with multiple random variables that are connected

- Nested plates
- Overlapping plates



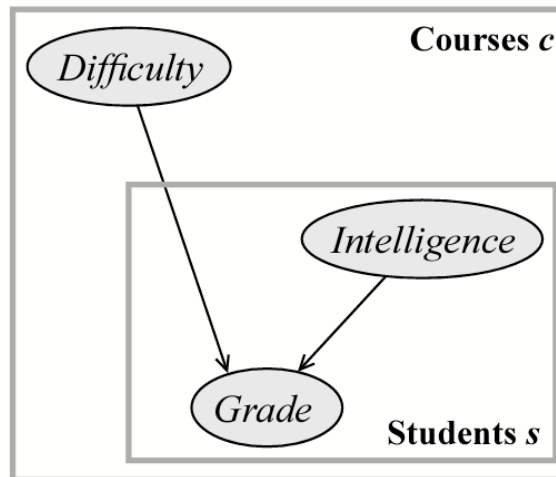
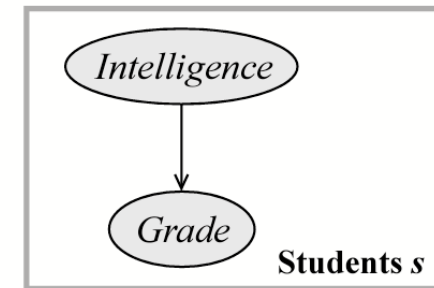
Nested plates

How to include difficulty of a course?

→ Adding the course plate.

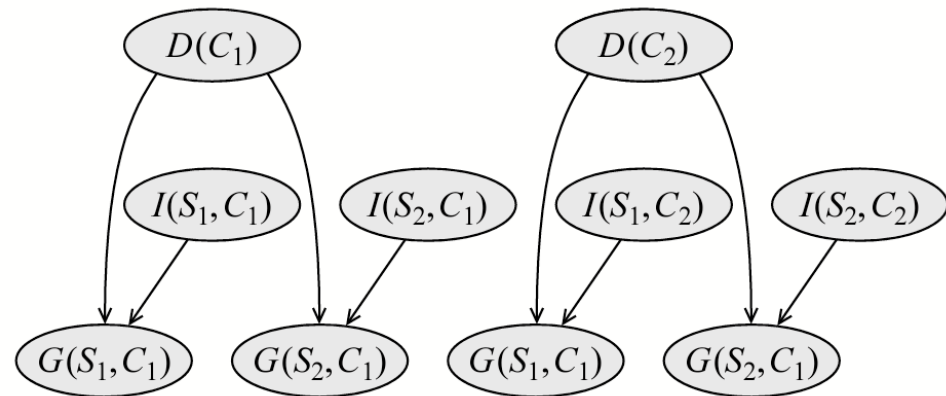
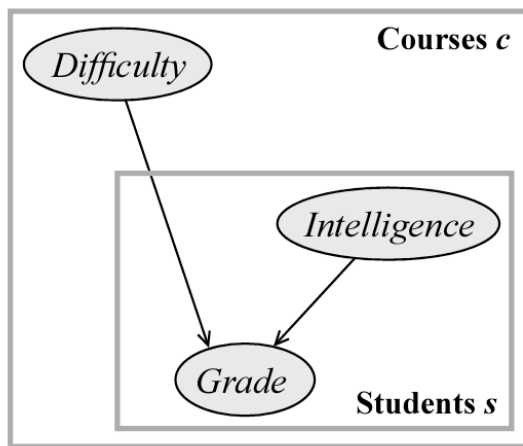
A course has several students.

→ Nested plates



Is it an appropriate model?

Nested plates

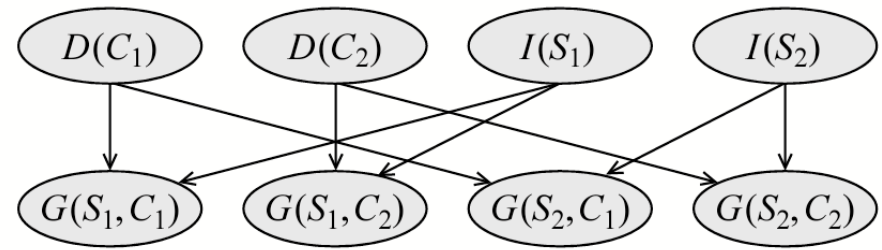
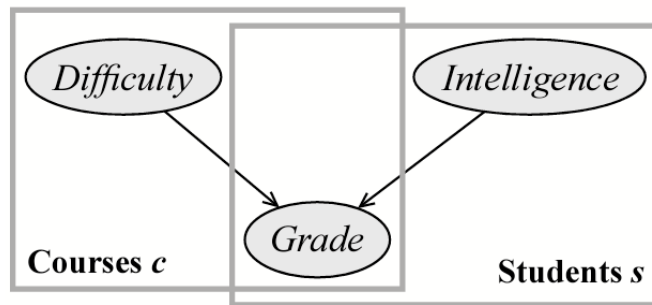


Course specific intelligence!

Intelligence is associated with student not with student and course.

→ Overlapping plates

Overlapping plates



D is the property of course
I is the property of student
Grade depends on both I and D

Our belief about a student's intelligence can be affected by the grades of other students in other classes

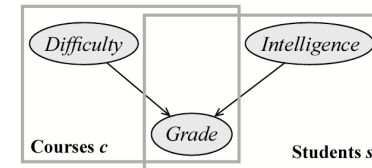
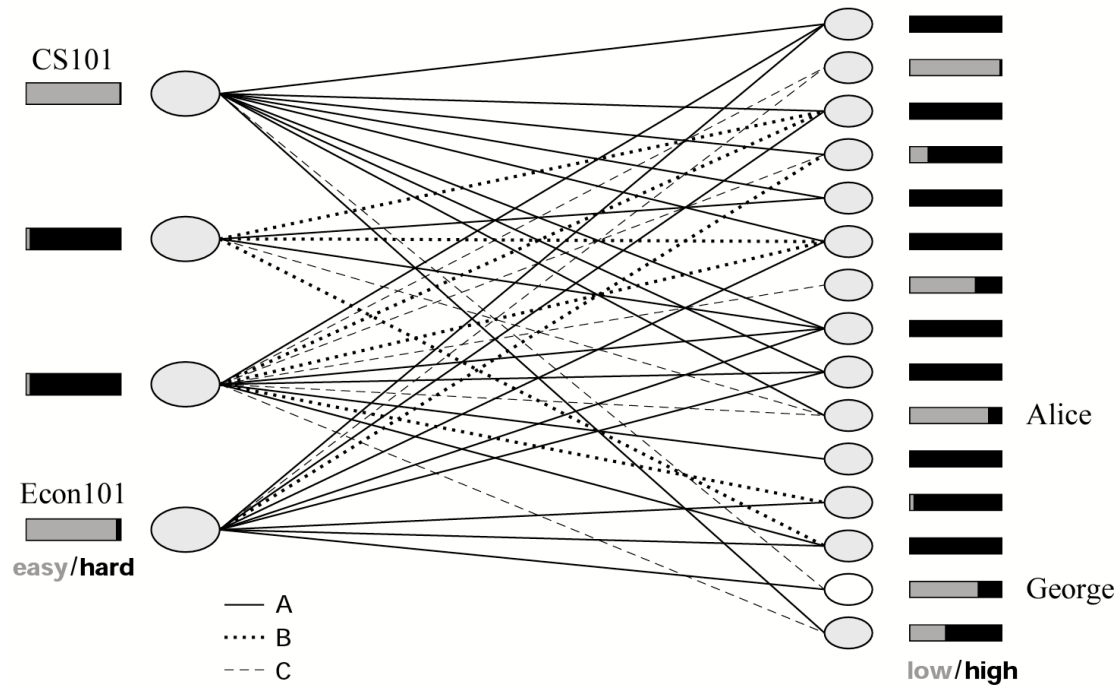
Power of Plate representations

- Even simple plate representations can induce fairly complex ground Bayesian networks
- Such networks model a rich network of interdependencies (correlation) between different variables
 - Allowing paths of influence that one may not anticipate

Modeling very rich correlated systems

Plate models encode correlations across multiple objects, allowing collective inference

- Collective Inference



Considering George's Grade in two courses. The probability of his intelligence is high. Considering entire body of evidence this probability decreases.

Conclusion is intuitive: Of all students who took CS101, only George got a C. Even Alice who got a C in both of her other classes got an A in CS101

Grade of other students affecting our belief about intelligence of a student

Plate Modeling Summary

- Representation and semantics are defined
- It uses the basic object-relational framework
- Plates correspond to object types

Plate dependency model

General description

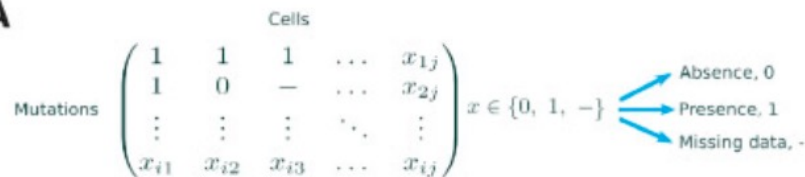
- A plate model M_{plate} defines for each template variable A with arguments $U_1, \dots, U_k$:
 - A set of template variable $A(U_1, \dots, U_k)$, e.g. $I(S_1, \dots, S_{100})$
 - Template parents $Pa_A = \{B_1(\mathbf{U}_1), \dots, B_m(\mathbf{U}_m)\}$, e.g. $G(\{C_i, S_j\})$
 - For each $B_i(\mathbf{U}_i)$ we must have $\mathbf{U}_i \subseteq U_1, \dots, U_k$
 - No indices in parent that are not in child $G(D, I)$
 - CPD $P(A \mid B_1, \dots, B_m)$
 - $P(G \mid D, I)$

Plate Model Example

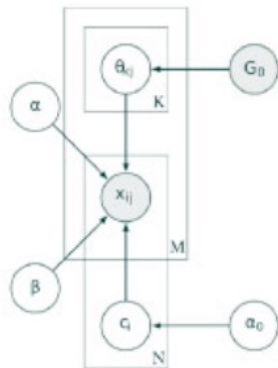
Sequence analysis

BnpC: Bayesian non-parametric clustering of single-cell mutation profiles

A



B



$$G_0 \equiv \text{Beta}(p, q)^M$$

$$\theta_k \sim G_0 \text{ for } k = 1, \dots, c$$

$$\alpha_0 \sim \Gamma(1, 1)$$

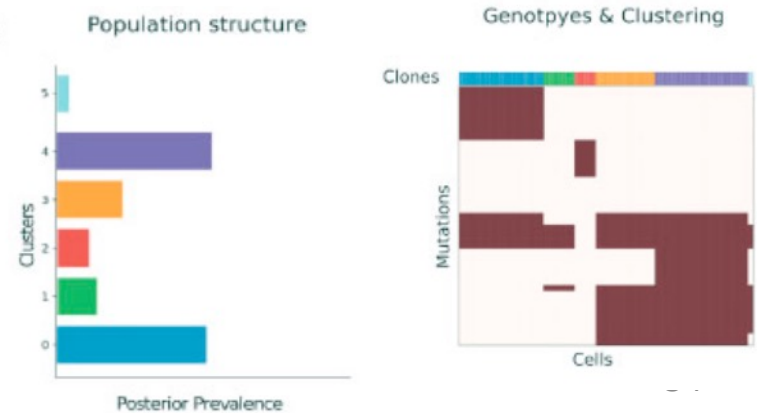
$$c_i \sim \text{CRP}(\alpha_0; c_1, \dots, c_{i-1})$$

$$\alpha \sim t\mathcal{N}(\mu_\alpha, \sigma_\alpha, 0, 1)$$

$$\beta \sim t\mathcal{N}(\mu_\beta, \sigma_\beta, 0, 1)$$

$$x_i \sim f(\theta_{c_i}, \alpha, \beta)$$

C

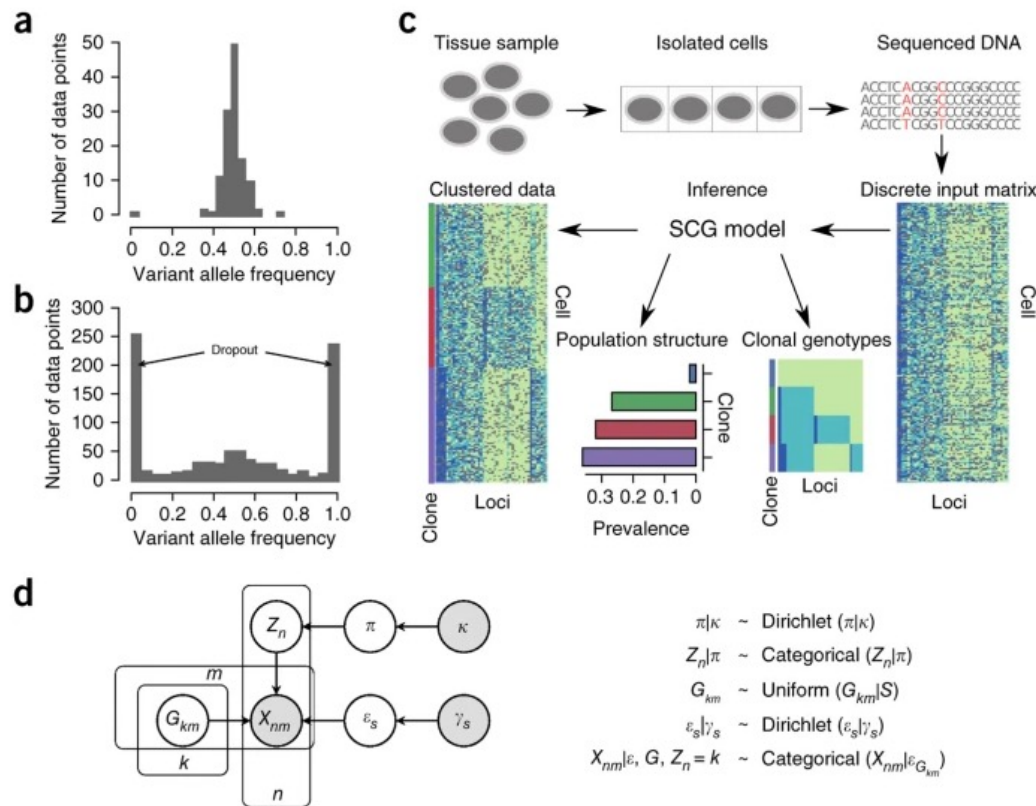


- Input data X: Binary matrix with N cells and M clones
- Noise: False positives (α) and false negatives (β)
- Genotypes: Base distribution G_0 over clone genotypes θ
- Assignments: Cells assigned to clones via CRP (concentration α)
- Likelihood: Links observed data to model parameters
- Inference: Unshaded nodes = inferred with MCMC

Plate Model Example

Nature methods

Clonal genotype and population structure inference from single-cell tumor sequencing



- Predict clonal genotypes, and infer the prevalence of clones
- a posterior distribution over the values of the unshaded nodes is approximated using a variational Bayesian method
 - π , clone prevalence;
 - Z_n , variable indicating clone of origin for cell n ;
 - G_{km} , variable indicating genotype of locus m for clone k ;
 - ϵ_s , error profile for genotype state s ;
 - X_{nm} , observed data from cell n and locus m ;
 - γ_s , parameter of Dirichlet distribution prior for the error profile for genotype state s ;
 - κ , parameter of Dirichlet distribution prior for the clone prevalence.